

JIAYU TANG

Gainesville, FL | jasper.tang859@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

Computational biotechnology researcher with antibody engineering and clinical-trial experience. **Skilled in:** AlphaFold-based structure prediction, statistical analysis, and cross-functional team leadership.

EDUCATION

2026 **M.S., Food Science and Technology; Minor in Statistics**

University of Nebraska–Lincoln

2024 **B.S., Food Science and Technology; B.Eng., Food Science and Engineering (Dual Degree)**

University of Nebraska–Lincoln / Northwest A&F University

RESEARCH EXPERIENCE

08/2026–Present **Research Assistant**

Department of Infectious Diseases and Immunology, University of Florida

- **Investigating** how antibody framework architecture affects antigen-binding prediction using fixed CDR, antigen, and epitope controls; built and evaluated an in-silico antibody-design pipeline with framework grafting and epitope-masked structural analysis.
- **Running** large-scale AlphaFold/ColabFold antibody–antigen predictions, evaluating model confidence, docking accuracy, and interface contacts; benchmarking emerging design tools including Germinal and RFantibody.

07/2024–07/2026 **Graduate Research Assistant**

Department of Food Science and Technology, University of Nebraska–Lincoln

- **Coordinated** a 12-week randomized crossover clinical trial (IRB-approved, ClinicalTrials.gov-registered) and a human biobank study; led 200+ clinic visits, recruitment, scheduling, and documentation.
- **Processed and analyzed** stool, blood, and dietary data using Python and R; co-developed an NIH-style protocol and authored 4 SOPs and 2 clinical workflows.
- **Led** a PICOS-based systematic review of 1,800+ references using AI-assisted (Gemini) and ML-based (ASReview, TERA-Tool) screening, validated against human reviewers.

SELECTED PUBLICATIONS

Published: Xiao J, Li Y, Niu L, et al. Foods. 2023;12(15):2964; Deng X, Niu L, Xiao J, et al. J Sci Food Agric. 2023;103(9):4257–4267.

SKILLS

Computational Biology: AlphaFold 3, ColabFold, Germinal, RFantibody; structural evaluation (RMSD, docking, epitope recovery)

Programming & Data Analysis: Python, R, SAS; experimental design, statistical modeling, reproducible data pipelines

Clinical & Research Operations: IRB/GCP-compliant clinical trials, biobanking, systematic reviews, REDCap, dataset QC; Mandarin (native), English (fluent)

LEADERSHIP

Graduate Representative, Graduate Student Assembly, UNL (2025–2026): Represent Food Science graduate students on academic-affairs issues in monthly university governance meetings.

Advisory Board Member, Dept. of Food Science & Technology, UNL (2024–2025): Launched a department-wide peer-mentoring program for incoming graduate students.

Communications Lead, Northwest A&F University (2020–2022): Led a 13-person team producing weekly promotional newsletters and video content for the college.